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Expandable element assembly with external rigid support element to isolate well interval

Abstract

An expandable element isolation assembly and a well interval isolation device utilizing at least a pair of expandable element isolation assemblies to isolate a well interval are disclosed. An expandable element isolation assembly can comprise an expandable element, and at least one rigid support element that is located external to the expandable element and is positionable between at least part of the expandable element and an inner wall of a formation at an open-hole portion of a wellbore in response to expansion of the expandable element. The at least one rigid support element can reinforce the expanded expandable element, such as through physical contact therewith. A retraction mechanism may be provided to retract the at least one rigid support element upon contraction of the expandable element.

Inventors: Reding; Christian (Houston, TX), Hudson; Marcus Ray (Pearland, TX), Zhang; Wei (Humble, TX)

Applicant: Halliburton Energy Services, Inc. (Houston, TX)

Family ID: 96174860

Assignee: HALLIBURTON ENERGY SERVICES, INC. (Houston, TX)

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Primary Examiner: Coy; Nicole

Assistant Examiner: Railey; Jennifer A

Attorney, Agent or Firm: Kilpatrick Townsend & Stockton LLP

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to hydrocarbon well operations, and more particularly although not necessarily exclusively, to an expandable element assembly for use in isolating well intervals.

BACKGROUND

(2) During a completion phase of a hydrocarbon well, an open hole well zone (or interval) of interest may be isolated from the remainder of a hydrocarbon well for various reasons. Isolating an open hole well interval of interest is commonly accomplished using a cooperating pair of spaced apart expandable elements, such as the inflatable packers of a straddle packer device. A straddle packer device can be positioned on a tool string that can be used to run the straddle packer to a desired downhole location. After the straddle packer is at the desired downhole location, the

expandable elements may be expanded to isolate the well interval located between the expandable elements from fluid and pressure in other portions of the well.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of a well interval isolation device located within an open hole portion of a hydrocarbon wellbore according to one example of the present disclosure.
- (2) FIG. 2 is a schematic representation of a well interval isolation device with expandable elements thereof in a contracted state according to one example of the present disclosure.
- (3) FIG. 3 is a schematic representation of a well interval isolation device with expandable elements thereof in an expanded state according to one example of the present disclosure.
- (4) FIG. 4 is an enlarged view depicting a rigid support element of a well interval isolation assembly in a closed position according to one example of the present disclosure.
- (5) FIG. 5 shows the rigid support element of FIG. 4 in an open position according to one example of the present disclosure.
- (6) FIG. 6 is a cross-section of the rigid support element shown in FIG. 5.
- (7) FIG. 7 is a flow chart representing a method for isolating an open hole interval of a hydrocarbon well according to one example of the present disclosure.

DETAILED DESCRIPTION

- (8) Certain aspects and examples of the present disclosure relate to an expandable element isolation assembly, and to a rigid support element that can increase a maximum differential pressure rating of the expandable element isolation assembly. By utilizing rigid support elements to limit or prevent undesirable deformation of the expandable isolation elements of an expandable element isolation assembly, the differential pressure between a well interval isolated by the expandable element isolation assembly and the hydrostatic pressure of the surrounding well fluid may be considerably greater than is normally possible through the use of known expandable elements.
- (9) An example of an expandable element isolation assembly can include an expandable element and an associated rigid support element. The rigid support element is located external to the expandable element and may be positionable between at least a part of the expandable element and an inner wall of a formation at an open-hole portion of a wellbore in response to expansion of the expandable element. When the rigid support element is so positioned upon expansion of the expandable element, the rigid support element can act as an external reinforcement of the expandable element. The reinforcement can substantially increase the differential pressure range with which the expandable element can be used.
- (10) Rigid support elements of various designs may be provided according to examples of an expandable element isolation assembly. For example, an annular array of overlapping rigid support petals may encircle one end of an expandable element according to one example of an expandable element isolation assembly. The rigid support petals may be positioned between at least part of the expandable element and an inner wall of a formation at an open-hole portion of a wellbore in response to expansion of the expandable element. Upon contraction of the expandable element, the rigid support petals may return to a retracted position where the rigid support petals move away from the inner wall of the wellbore and toward a central axis or centerline of the wellbore.
- (11) In at least one example of an expandable element isolation assembly, the rigid support petals or another rigid support element may be retracted by a spring acting on a linkage coupled to the rigid support petals/element. In another example, the rigid support petals or another rigid support element may be directly retracted by a linear actuator, or an elastomeric ring or band. The use of other retraction mechanisms is also possible.
- (12) An expandable element isolation assembly may be a part of a well interval isolation device

that is positionable on a wireline tool string or a tubing string for deploying the well interval isolation device to a target depth/location within a wellbore. A pair of expandable element isolation assemblies may reside in a spaced apart relationship at opposite ends of a tool portion of the well interval isolation device. The distance between the expandable element isolation assemblies effectively defines the length of the well interval that can be isolated by the well interval isolation device. Different types of tools can be used. For example, a well interval isolation device may utilize a tool via which fluid in an isolated well interval can be extracted for analysis or the pressure within the isolated well interval can otherwise be reduced by evacuation. In another example, a well interval isolation device may utilize a tool via which pressurized fluid can be expelled into an area of the formation that is located within the isolated well interval, such as for purposes of creating microfractures in the formation.

(13) In operation, a well interval isolation device can be delivered into a wellbore to a target depth, such as to a target depth that coincides with a well interval of interest. Upon reaching the target depth, the expandable elements of each expandable element isolation assembly can be expanded, such as through inflation by pressurized fluid. Expansion of the expandable elements can create a seal with the rock or other formation material forming the wellbore wall and can thereby isolate the well interval located between the expandable elements from the fluid in the remainder of the wellbore. The rigid support element respectively associated with each of the expandable elements can be automatically extended via expansion of the expandable elements, as described above. The reinforcement provided by the rigid support elements can help the expandable elements resist deformation and maintain a seal with the wall of the wellbore at higher pressure differentials than would otherwise be possible using expandable elements of known design.

(14) Illustrative examples follow and are given to introduce the reader to the general subject matter discussed herein rather than to limit the scope of the disclosed concepts. The following sections describe various additional features and examples with reference to the drawings in which like numerals indicate like elements, and directional descriptions are used to describe the illustrative aspects, but, like the illustrative aspects, should not be used to limit the present disclosure.

(15) One example of a well interval isolation device **200** being used to isolate an interval **150** of an open hole portion of a hydrocarbon well **100** is represented in the diagram of FIG. **1**. As shown, the hydrocarbon well **100** can include a rig **102** located at a well surface **104**. The rig can be used to convey the well interval isolation device **200** relative to a wellbore **106**. The wellbore **106** in FIG. **1** is drilled into a subterranean formation **108**. In other examples, a wellbore can be drilled through a sub-sea formation. The wellbore **106** is shown to include a vertical portion **110**, as well as a horizontal portion **112** that may be absent in other examples. In some examples, the wellbore **106** may also include a deviated portion. The well may be a hydraulic fracturing well. While the entirety of the wellbore **106** is depicted in FIG. **1** as an open-hole wellbore for purposes of illustration, it should be understood that in other examples, at least a portion of the wellbore **106** may have a casing installed therein.

(16) In the example shown, the well interval isolation device **200** is part of a tool string attached to tubing **114**, which may be coiled tubing. In another example, the well interval isolation device **200** may instead be part of a tool string that is conveyed by a wireline. While the rig **102** of the hydrocarbon well **100** is used to convey the well interval isolation device **200** according to the example of FIG. **1**, in other examples, the rig **102** may be replaced with a wellhead (e.g., Christmas tree) or another well completion apparatus, and a wireline rigup may be present to convey the well interval isolation device **200** relative to the wellbore **106**.

(17) During a wellbore drilling operation, drilling fluid (“mud”) can be pumped downhole to cool a drill bit. During the wellbore drilling operation, the wellbore **106** may also partially fill with various fluids that seep into the wellbore **106** from the formation **108**. Samples of the fluids present in the wellbore **106** may be extracted for analysis, such as to determine the nature and desirability (e.g., quality) of the fluids present in various portions of the formation **108**. Zonal isolation—i.e.,

isolation of a particular well interval—may be used in this regard to help ensure that a fluid sample of interest can be extracted without also extracting nearby fluids, or an excessive volume of nearby fluids.

(18) When the well is a hydraulic fracturing well, the formation is typically fractured in various locations using high pressure fluid. To optimize fracture placement and for other reasons, well operators may utilize microfracturing techniques to measure stresses within different portions of the formation. Microfracturing techniques commonly utilize one or more jets of highly pressurized fluid to create small fractures in the formation **108**. As is the case with fluid sampling/extraction, a microfracturing operation may be performed within an isolated well interval.

(19) FIGS. **2-3** each depict the well interval isolation device **200** example shown in FIG. **1**. As shown, the well interval isolation device **200** can include at least one pair of expandable element isolation assemblies comprising an uphole expandable element isolation assembly **210** and a downhole expandable element isolation assembly **215**. A tool **220**, such as a fluid extraction tool or a microfracturing tool, can reside between the uphole expandable element isolation assembly **210** and the downhole expandable element isolation assembly **215**. Various valves **225** and/or other components of the well interval isolation device **200** may be located between a downhole end of the downhole expandable element isolation assembly **215** and a remainder of the tool string **205**, as shown.

(20) Each of the uphole expandable element isolation assembly **210** and the downhole expandable element isolation assembly **215** is shown to include an expandable element **230** and an associated rigid support element **235**. The uphole expandable element isolation assembly **210** and the downhole expandable element isolation assembly **215** are arranged in a mirrored relationship on the tool string **205** in this example.

(21) In FIG. **2**, the expandable elements **230** of each of the uphole expandable element isolation assembly **210** and the downhole expandable element isolation assembly **215** are shown in a contracted state. In FIG. **3**, each of the expandable elements **230** of the uphole expandable element isolation assembly **210** and the downhole expandable element isolation assembly **215** are expanded, as they would be when the well interval isolation device **200** is used to isolate an interval of an open-hole wellbore. The expandable elements **230** can be expanded in a variety of ways. For example, when the expandable elements **230** are inflatable expandable elements, such as inflatable packers, the expandable elements **230** may be inflated using pressurized fluid. In another example, the expandable elements **230** may instead be comprised of a deformable material such as an elastomeric material. When comprised of a deformable material, the expandable elements **230** may be dynamically expanded by, for example, compression caused by linear translation of one or more of the tool string **205** components.

(22) In an example where the expandable elements **230** are inflatable packers or similar inflatable elements, pressurized fluid may be supplied to the expandable elements **230** from a source of pressurized fluid. The source of pressurized fluid can be located at the well surface. The pressurized fluid may be transmitted to the expandable elements **230** using a pump or a similar apparatus. One or more pressurized fluid conduits may run along an open interior of the tool string **205** to connect one or more sources of pressurized fluid located at the well surface with each of the expandable elements **230**. The one or more sources of pressurized fluid may be connected to the expandable elements **230** in various configurations, such that the expandable elements **230** may be inflated individually or simultaneously. According to one example, the expandable elements **230** can be individually inflated, either through a unique connection between each expandable element **230** and the one or more sources of pressurized fluid, or by utilizing one or more valves to selectively distribute the pressurized fluid to the individual expandable elements **230**. In another example, the expandable elements **230** may be connected to the one or more sources of pressurized fluid in a manner such that the expandable elements **230** can only be simultaneously inflated.

(23) Referring again to FIG. **1**, it may be observed that the expandable elements **230** of the well

interval isolation device **200** are expanded (e.g., inflated) such that an exterior surface thereof is in sealing contact with the rock or other formation material forming the wall of the wellbore **106**. The expandable elements **230** can thereby isolate the interval **150** of the wellbore **106** residing between the expandable elements **230** from pressure and other fluid in the wellbore **106**. With the well interval isolation device **200** isolating the well interval **150**, the tool **220** of the well interval isolation device **200** can be used to perform a desired operation within the isolated well interval **150**. As described above, the operation may be a fluid extraction (drawdown) operation or an overpressure (e.g., microfracturing) operation in some examples. Once the desired operation within the well interval **150** is complete, the expandable elements **230** can be contracted (e.g., deflated) and the well interval isolation device **200** may be moved to another wellbore location of interest. The expandable elements **230** can then be reinflated to isolate another well interval.

(24) One example of the uphole expandable element isolation assembly **210** of FIGS. 2-3 may be observed in more detail in FIGS. 4-6. As briefly described above, and as shown in FIGS. 4-6, the expandable element isolation assembly **215** according to this example may include the expandable element **230** in combination with the rigid support element **235**. While only the uphole expandable element isolation assembly **210** is described in more detail below for purposes of simplicity, it should be understood that the design, construction, and operation of the uphole expandable element isolation assembly **210** and the downhole expandable element isolation assembly **215** may be the same.

(25) FIG. 4 depicts the uphole expandable element **230** of the uphole expandable element isolation assembly **210** as an inflatable packer in a deflated state. FIGS. 5-6 depict the expandable element **230** of the uphole expandable element isolation assembly **210** as an inflatable packer in an inflated state. It may be observed from FIGS. 4-6 that the rigid support element **235** is located externally of the expandable element **230** of the uphole expandable element isolation assembly **210**. The same component arrangement would exist for the expandable element **230** of the downhole expandable element isolation assembly **210**.

(26) Expansion of the expandable elements **230** can create a seal with the rock or other formation material forming the wellbore wall and can thereby isolate the well interval located between the expandable elements from the fluid in the remainder of the wellbore, as is generally illustrated in FIG. 1. As may be best understood by reference to FIGS. 5-6, the rigid support element **235** is positionable between at least part of the expandable element **230** and the wall of the formation **108** that forms the open-hole portion of the wellbore **106** in response to expansion of the expandable element **230**. That is, outward movement of a portion of the rigid support element **235** toward the wall of the wellbore **106** may be caused by expansion of the expandable element **230**.

(27) The expandable element **230** and the rigid support element **235** may be coupled to a body **240** of the expandable element isolation assembly **210**. The body **240** of the expandable element isolation assembly **210** may be coupled to a downhole tool, (e.g., tool **220**) by an adapter **245**. The adapter **245** may also couple the expandable element isolation assembly **210** to the tool string **205** or to another conveyance via which the expandable element isolation assembly **210** can be moved along the length of a wellbore. The body **240** and the adapter **245** are preferably rigid components.

(28) The rigid support element **235** may be of various designs in different examples. In the particular example shown in FIGS. 4-6, the rigid support element **235** includes an assembly comprising an annular array of overlapping and pivotable rigid support petals **250** that encircle the uphole end of the expandable element **230**. Each rigid support petal **250** can be pivotably connected at a first end thereof to the body **240** by a fixed position coupling element **255**. The rigid support petals **250** may be positioned between at least part of the expandable element and the inner wall of the formation **108** at an open-hole portion of the wellbore **106** in response to expansion of the expandable element **230** (see FIGS. 5-6). Upon contraction/deflation of the expandable element **230**, the rigid support petals **250** may return to a retracted position where the rigid support petals **250** are moved away from the inner wall of the wellbore **106** and toward the body **240** of the

expandable element isolation assembly **210** and a centerline C/L of the wellbore **106** (see FIG. 4). (29) According to this example of the expandable element isolation assembly **210**, return of the rigid support petals **250** to the retracted position depicted in FIG. 4 can be caused by a retraction mechanism **260** subsequent to or in conjunction with deflation of the expandable element **230**. One example of the retraction mechanism **260** may include a plurality of linkage members **265** that are pivotably connected at first ends thereof to second ends of corresponding ones of the rigid support petals **250**. An opposite, second end of each linkage member **265** may be pivotably connected to a translating element **270** that is rotationally constrained but free to move linearly within some limited distance along the length of the body **240**. The adapter **245** may serve as a fixed-position retainer element, or a separate fixed-position retainer element **275** may be located along the downhole end of the adapter **245** and uphole of the translating element **270**, such that a space exists therebetween.

(30) The translating element **270**, and the adapter **245** or the retainer element **275**, may act to cooperatively retain a compression spring **280** or another resilient element that exerts a biasing force against the translating element **270**. The force exerted on the rigid support petals **250** during inflation of the expandable element **230** can be transferred to the linkage members **265** to cause a linear displacement of the translating element **270** toward the adapter **245** by overcoming the biasing force of the compression spring **280**. Upon deflation of the expandable element **230**, the biasing force of the compression spring **280** can produce an opposite direction linear displacement of the translating element **270** toward the expandable element **230**. This opposite direction linear displacement of the translating element **270** can also cause the linkage members **265** to rotate the rigid support petals **250** inward toward the deflated expandable element **230** and the tool **220**. The inwardly-directed rotation of the rigid support petals **250** allows the expandable element isolation assembly **210** to maintain a minimal profile, which facilitates movement of the well interval isolation device **200** through the wellbore **106**.

(31) It should be understood that the arrangement and orientation of the expandable element isolation assembly **210** in the particular example illustrated by FIGS. 4-6 represents a case where the tool **220** of the well interval isolation device **200** will be used to perform a pressure-increasing operation within an isolated well interval located between the expandable elements **230**. One example of such an isolated well interval **150** is shown in FIG. 1. In such a case, increased pressure within the isolated interval can produce a force that is directed away from the tool **220** in both an uphole and a downhole direction. The uphole-directed force will encourage deformation of the uphole expandable element **230** in an uphole direction, while the downhole-directed force will encourage deformation of the downhole expandable element **230** in a downhole direction—i.e., both expandable elements will be biased away from the tool **220**.

(32) To counter this encouraged deformation of the expandable elements **230**, the uphole expandable element isolation assembly **210** can be arranged on the uphole side of the uphole expandable element **230**. While not shown in FIGS. 4-6, the downhole expandable element isolation assembly **215** can likewise be arranged on the downhole side of the downhole expandable element **230**. Additionally, and as may be best observed in FIGS. 5-6, the uphole expandable element isolation assembly **210** can be oriented with the rigid support petals **250** thereof adjacent to an uphole end of the uphole expandable element **230**. While not shown, the downhole expandable element isolation assembly **215** can be oriented as a mirror image to the uphole expandable element isolation assembly **210**, such that the rigid support petals **250** of the downhole expandable element isolation assembly **215** are adjacent to a downhole end of the downhole expandable element **230**. As such, expansion of the expandable elements **230** will position the rigid support petals **250** of the associated expandable element isolation assemblies **210**, **215** between at least part of the corresponding expandable elements **230** and an inner wall of a formation, and resulting contact of the rigid support petals **250** with the expandable elements **230** can minimize or prevent deformation of the expandable elements **230** away from the tool **220** due to increased pressure in

the isolated interval, or otherwise.

(33) In an alternative example, the tool **220** of the well interval isolation device **200** may be used to perform a pressure-decreasing (e.g., drawdown) operation within an isolated well interval located between the expandable elements **230**. In such a case, decreased pressure within the isolated interval can produce a force that is directed toward the tool **220** from both an uphole and a downhole direction. These forces will encourage deformation of the uphole expandable element **230** in a downhole direction and deformation of the downhole expandable element **230** in an uphole direction—i.e., both expandable elements will be biased toward the tool **220**.

(34) To counter this encouraged deformation of the expandable elements **230**, the uphole expandable element isolation assembly **210** can be arranged on the downhole side of the uphole expandable element **230**. The downhole expandable element isolation assembly **215** can likewise be arranged on the uphole side of the downhole expandable element **230**. Additionally, the uphole expandable element isolation assembly **210** can be oriented with the rigid support petals **250** thereof adjacent to a downhole end of the uphole expandable element **230**. The downhole expandable element isolation assembly **215** can be oriented as a mirror image to the uphole expandable element isolation assembly **210**, such that the rigid support petals **250** of the downhole expandable element isolation assembly **215** are adjacent to an uphole end of the downhole expandable element **230**. As such, expansion of the expandable elements **230** will position the rigid support petals **250** of the associated expandable element isolation assemblies **210**, **215** between at least part of the expandable elements **230** and an inner wall of a formation, and resulting contact of the rigid support petals **250** with the expandable elements **230** can minimize or prevent deformation of the expandable elements **230** toward the tool **220** due to a decreased pressure in the isolated interval, or otherwise.

(35) In operation, the well interval isolation device **200** can be placed at a target location (e.g., target depth) within the wellbore **106**. In the example of FIG. 1, the target location is an open-hole portion of the wellbore **106**. After being positioned in the wellbore **106**, the expandable elements **230** of the expandable element isolation assemblies **210**, **215** can be expanded until an outside surface of the expandable elements makes sealing contact with the wall of the wellbore **106** (i.e., with the formation **108** in the case of an open hole portion of the wellbore **106**). Inflation of the expandable elements **230** causes the rigid support petals **250** of the respective expandable element isolation assemblies **210**, **215** to rotate outward and to become positioned between at least part of the expandable elements and the rock or other material of the formation **108** that forms the inner wall of the wellbore **106**. The rigid support petals **250**, via their location relative to the expandable elements **230** and their connection to the rigid bodies **240** of the respective expandable element isolation assemblies **210**, **215**, can minimize extrusion gaps between the expandable elements **230** and the formation **108** and reinforce and support the expandable elements **230** through physical contact therewith. The pressure differential rating of the expandable elements **230** may thus be increased by the use of the rigid support petals **250** or other rigid support elements. The increased pressure differential rating can allow a well interval isolated by the well interval isolation device **200** to be subjected to a greater (negative) drawdown pressure or (positive) overpressure than would otherwise be possible when using non-reinforced expandable elements.

(36) As may be observed in FIGS. 5-6, portions/surfaces of the rigid support petals **250** of the uphole expandable element isolation assembly **210** can be placed into contact with the rock or other material of the formation **108** that forms the wall of the wellbore **106** by expansion of the expandable element **230**. The wall of the wellbore **106** also acts as a hard stop to further rotation of the rigid support petals **250**, and enhances the support and resistance to deformation imparted to the uphole expandable element **230** by the uphole expandable element isolation assembly **210** and the rigid support petals **250** thereof. A like interaction between the rigid support petals **250** of the downhole expandable element isolation assembly **215** and the wall of the wellbore **106** similarly enhances the support and resistance to deformation imparted to the downhole expandable element

230 by the downhole expandable element isolation assembly **215** and the rigid support petals **250** thereof.

(37) According to some examples, an expandable element isolation assembly, such as a packer assembly, can have a design and construction that differs from the design and construction of the expandable element isolation assembly **210** shown in FIGS. **4-6**. For example, in one alternative example, the inflatable expandable element **230** can be replaced with an expandable (e.g., elastic) element of substantially solid elastomeric or similar construction. In such an example, the substantially solid expandable element may be expanded radially outward through forced deformation, such as by compressing the expandable element by a linear actuator or by translating movement of one or more components of a tool string of which the expandable element isolation assembly is a part.

(38) In another example, at least the compression spring **280**, and the translating element **270** of the expandable element isolation assembly **210** of FIGS. **4-6** can be omitted. The linkage members **265** can also be replaced with one or more motive devices such as compression springs, leaf springs, or linear actuators that are directly pivotably coupled between the body **240** and the second ends of the rigid support petals **250** to produce the inwardly-directed retraction thereof. In another example, at least the compression spring **280**, the translating element **270**, and the linkage members **265** of the expandable element isolation assembly **210** of FIGS. **4-6** may be replaced with powered rotary actuators, torsion springs, or other rotation inducing motive devices that are coupled to the rigid support petals **250**. Such rotation-inducing motive devices can directly rotate the rigid support petals **250** in an inward direction about the first ends thereof upon contraction/deflation of the expandable element **230**. In still another example, at least the compression spring **280**, the translating element **270**, and the linkage members **265** of the expandable element isolation assembly **210** of FIGS. **4-6** may be replaced with an annular spring, or with an elastomeric band or ring, such as an O-ring, which encircles and is in contact with the rigid support petals **250**. In such an example, the contracting force of the elastomeric band/ring or annular spring can be overcome by the expansion force of the expandable element **230**, but the elastomeric band/ring or annular spring can have sufficient contraction strength to rotate the rigid support petals **250** in an inward direction about the first ends thereof upon contraction (e.g., deflation) of the expandable element **230**.

(39) FIG. **7** is a flowchart representing a process for isolating an interval of an open-hole portion of a wellbore using a well interval isolation device **200** having expandable element isolation assemblies. As represented in FIG. **6**, at block **300** a well interval isolation device according to an example of the present disclosure can be positioned at a target location within an open-hole portion of a wellbore. As described above, the well interval isolation device may comprise a downhole tool in conjunction with an uphole expandable element isolation assembly and a downhole expandable element isolation assembly that are coupled to the downhole tool near opposite ends thereof. According to one example, each expandable element isolation assembly may comprise an expandable element and at least one rigid support element located external to the expandable element. The at least one rigid support element may be positionable between at least part of the expandable element and an inner wall of a formation at an open-hole portion of the wellbore in response to expansion of the expandable element. Each expandable element isolation assembly may also include a retraction mechanism for retracting the at least one rigid support element upon contraction of the expandable element.

(40) At block **302**, each of the expandable elements of the expandable element isolation assemblies of the well interval isolation device can be expanded. Expansion of the expandable elements cause an outside surface of each expandable element to create a seal with an inner wall of the formation. In an example, the expandable elements can be inflated using pressurized fluid, such as pressurized fluid supplied by a pump from a fluid source located at the well surface. The pressure within the expandable elements may be monitored during the expandable element inflation process.

(41) In block **304**, the at least one rigid support element associated with each expandable element can be positioned between at least part of the expandable element and the inner wall of the formation in response to expansion of the expandable elements. For example, in the case of the expandable element isolation assembly **210** shown in FIGS. **4-6**, expansion of an expandable element **230** causes the rigid support petals **250** of the associated rigid support element **235** to be positioned between a portion of the expandable element **230** and an inner wall of the formation **108**. With the well interval of interest isolated from surrounding pressure and well fluids by the reinforced expanded expandable elements, the downhole tool can be used to perform an operation within the well interval. The operation may be, without limitation, a drawdown operation that extracts fluid from the interval and reduces the pressure thereof, or an overpressure operation such as a microfracturing operation that increases the pressure within the well interval. Other operations are also possible.

(42) When it is desired to remove the well interval isolation device from the wellbore or to relocate the well interval isolation device within the wellbore, the expandable elements may be contracted, as indicated at block **306**. According to the example process represented in FIG. **6**, the expandable elements may be contracted by extracting pressurized fluid from within the expandable elements. For example, the aforementioned pump may be used to evacuate fluid from within the expandable elements.

(43) As represented at block **308**, with the expandable elements contracted or in the process of contracting, the rigid support elements may be retracted by action of a retraction mechanism associated with each expandable element isolation assembly. For example, in the case of the expandable element isolation assembly **210** shown in FIGS. **4-6**, the biasing force of the compression spring **280** causes a linear displacement of the translating element **270** toward the expandable element **230**, which causes the linkage members **265** to rotate the rigid support petals **250** inward toward the deflating/deflated expandable element **230** and the tool **220**. The well interval isolation device may then be relocated within the wellbore or removed from the wellbore as indicated at block **310**. For example, the tool string on which the well interval isolation device is positioned, can be moved in an uphole or downhole direction relative to the wellbore. This allows the well interval isolation device to be repositioned to a new wellbore target location, whereafter the well interval isolation device can again be used to isolate a desired well interval.

(44) According to aspects of the present disclosure, an expandable element isolation assembly, a well interval isolation device, and a method of isolating an interval of an open-hole portion of a wellbore, are provided according to one or more of the following examples. As used below, any reference to a series of examples is to be understood as a reference to each of those examples disjunctively (e.g., “Examples 1-4” is to be understood as “Examples 1, 2, 3, or 4”).

(45) Example 1 is an expandable element isolation assembly for use in a wellbore, the expandable element isolation assembly comprising: an expandable element; and at least one rigid support element located external to the expandable element and positionable between at least part of the expandable element and an inner wall of a formation at an open-hole portion of the wellbore in response to expansion of the expandable element.

(46) Example 2 is the expandable element isolation assembly of example 1, wherein the expandable element is an inflatable packer.

(47) Example 3 is the expandable element isolation assembly of example 1, wherein the at least one rigid support element is an assembly comprising an annular array of overlapping and pivotable rigid support petals that encircle one end of the expandable element.

(48) Example 4 is the expandable element isolation assembly of example 3, wherein each rigid support petal is pivotably connected at a first end thereof to a body of a downhole tool to which the expandable element isolation assembly is coupled.

(49) Example 5 is the expandable element isolation assembly of example 4, further comprising a retraction mechanism for returning the rigid support petals to a retracted position upon contraction

of the expandable element.

(50) Example 6 is the expandable element isolation assembly of example 5, wherein the retraction mechanism comprises: a plurality of linkage members pivotably connected at first ends thereof to second ends of corresponding ones of the rigid support petals, second ends of the plurality of linkage members pivotably connected to a translating element that is coupled to the body so as to be linearly displaceable relative thereto; and a spring exerting a biasing force against the translating element in a direction of the expandable element; wherein, upon contraction of the expandable element, the biasing force of the spring is directed to cause a linear displacement of the translating element toward the expandable element and a resulting inward rotation of the rigid support petals by the linkage members.

(51) Example 7 is the expandable element isolation assembly of example 5, wherein the retraction mechanism comprises a motive device selected from the group consisting of a compression spring, a leaf spring, and a linear actuator, the motive device directly pivotably coupled between the body and second ends of the rigid support petals.

(52) Example 8 is the expandable element isolation assembly of example 5, wherein the retraction mechanism encircles and is in contact with the rigid support petals and comprises a device selected from the group consisting of an elastomeric band, an elastomeric ring, and an annular spring.

(53) Example 9 is a well interval isolation device comprising: an uphole expandable element isolation assembly and a downhole expandable element isolation assembly coupled to a downhole tool near opposite ends thereof, each expandable element isolation assembly comprising: an expandable element; and at least one rigid support element located external to the expandable element and positionable between at least part of the expandable element and an inner wall of a formation at an open-hole portion of a wellbore in response to expansion of the expandable element.

(54) Example 10 is the well interval isolation device of example 9, wherein each expandable element is an inflatable packer.

(55) Example 11 is the well interval isolation device of example 9, wherein the at least one rigid support element is an assembly comprising an annular array of overlapping and pivotable rigid support petals that encircle one end of each expandable element.

(56) Example 12 is the well interval isolation device of example 11, further comprising a retraction mechanism for returning the rigid support petals to a retracted position upon deflation of the expandable elements.

(57) Example 13 is the well interval isolation device of example 12, wherein the retraction mechanism comprises: a plurality of linkage members pivotably connected at first ends thereof to second ends of corresponding ones of the rigid support petals, second ends of the plurality of linkage members pivotably connected to a translating element that is coupled to a body so as to be linearly displaceable relative thereto; and a spring exerting a biasing force against the translating element in a direction of the expandable element; wherein, upon contraction of the expandable element, the biasing force of the spring is directed to cause a linear displacement of the translating element toward the expandable element and a resulting inward rotation of the rigid support petals by the linkage members.

(58) Example 14 is the well interval isolation device of example 12, wherein the retraction mechanism comprises a motive device selected from the group consisting of a compression spring, a leaf spring, and a linear actuator, the motive device directly pivotably coupled between the body and second ends of the rigid support petals.

(59) Example 15 is the well interval isolation device of example 12, wherein the retraction mechanism encircles and is in contact with the rigid support petals and comprises a device selected from the group consisting of an elastomeric band, an elastomeric ring, and an annular spring.

(60) Example 16 is a method comprising: positioning a well interval isolation device at a target location within an open-hole portion of a wellbore, the well interval isolation device comprising: a

downhole tool; and an uphole expandable element isolation assembly and a downhole expandable element isolation assembly coupled to the downhole tool near opposite ends thereof, each expandable element isolation assembly comprising: an expandable element; at least one rigid support element located external to the expandable element and positionable between at least part of the expandable element and an inner wall of a formation at an open-hole portion of the wellbore in response to expansion of the expandable element; and a retraction mechanism for retracting the at least one rigid support element upon contraction of the expandable element; and isolating an interval of the wellbore by expanding the expandable elements such that an outside surface of each expandable element is forced into contact with the inner wall of the formation and the at least one rigid support element associated with each expandable element is positioned between at least part of the expandable element and the inner wall of the formation by expansion of the expandable element.

(61) Example 17 is the method of example 16, wherein the retraction mechanism comprises: a plurality of linkage members pivotably connected at first ends thereof to second ends of corresponding ones of an annular array of overlapping and pivotable rigid support petals, second ends of the plurality of linkage members pivotably connected to a translating element that is coupled to a body so as to be linearly displaceable relative thereto; and a spring exerting a biasing force against the translating element in a direction of the expandable element; wherein, upon contraction of the expandable element, the biasing force of the spring causes a linear displacement of the translating element toward the expandable element, which produces an inward rotation of the rigid support petals by the linkage members.

(62) Example 18 is the method of example 17, wherein the retraction mechanism comprises a motive device selected from the group consisting of a compression spring, a leaf spring, and a linear actuator, the motive device directly pivotably coupled between the body and second ends of the rigid support petals.

(63) Example 19 is the method of example 17, wherein the retraction mechanism encircles and is in contact with the rigid support petals and comprises a device selected from the group consisting of an elastomeric band, an elastomeric ring, and an annular spring.

(64) Example 20 is the method of example 16, wherein the expandable elements are expanded by inflation with a pressurized fluid.

(65) The foregoing description of certain examples, including illustrated examples, has been presented only for the purpose of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Numerous modifications, adaptations, and uses thereof will be apparent to those skilled in the art without departing from the scope of the disclosure.

Claims

1. An expandable element isolation assembly for use in a wellbore, the expandable element isolation assembly comprising: an expandable element; a rigid support element comprising an assembly of an annular array of overlapping and pivotable rigid support petals, each of which is pivotably connected at a first end thereof to a body of a downhole tool to which the expandable element isolation assembly is coupled, the annular array of overlapping and pivotable rigid support petals encircling one end of the expandable element such that a portion of the annular array of overlapping and pivotable rigid support petals is positionable between the one end of the expandable element and an inner wall of a formation at an open-hole portion of the wellbore in response to expansion of the expandable element; and a retraction mechanism for returning the rigid support petals to a retracted position upon contraction of the expandable element.
2. The expandable element isolation assembly of claim 1, wherein the expandable element is an inflatable packer.

3. The expandable element isolation assembly of claim 1, wherein the retraction mechanism comprises: a plurality of linkage members pivotably connected at first ends thereof to second ends of corresponding ones of the rigid support petals, second ends of the plurality of linkage members pivotably connected to a translating element that is coupled to the body so as to be linearly displaceable relative thereto; and a spring exerting a biasing force against the translating element in a direction of the expandable element; wherein, upon contraction of the expandable element, the biasing force of the spring is directed to cause a linear displacement of the translating element toward the expandable element and a resulting inward rotation of the rigid support petals by the linkage members.
4. The expandable element isolation assembly of claim 1, wherein the retraction mechanism comprises a motive device selected from the group consisting of a compression spring, a leaf spring, and a linear actuator, the motive device directly pivotably coupled between the body and second ends of the rigid support petals.
5. The expandable element isolation assembly of claim 1, wherein the retraction mechanism encircles and is in contact with the rigid support petals and comprises a device selected from the group consisting of an elastomeric band, an elastomeric ring, and an annular spring.
6. A well interval isolation device comprising: an uphole expandable element isolation assembly and a downhole expandable element isolation assembly coupled to a downhole tool near opposite ends thereof, each expandable element isolation assembly comprising: an expandable element; a rigid support element comprising an assembly of an annular array of overlapping and pivotable rigid support petals, each of which is pivotably connected at a first end thereof to a body of a downhole tool to which the expandable element isolation assembly is coupled, the annular array of overlapping and pivotable rigid support petals encircling one end of the expandable element such that a portion of the annular array of overlapping and pivotable rigid support petals is positionable between the one end of the expandable element and an inner wall of a formation at an open-hole portion of a wellbore in response to expansion of the expandable element; and a retraction mechanism for returning the rigid support petals to a retracted position upon contraction of the expandable element.
7. The well interval isolation device of claim 6, wherein each expandable element is an inflatable packer.
8. The well interval isolation device of claim 6, wherein the retraction mechanism comprises: a plurality of linkage members pivotably connected at first ends thereof to second ends of corresponding ones of the rigid support petals, second ends of the plurality of linkage members pivotably connected to a translating element that is coupled to a body so as to be linearly displaceable relative thereto; and a spring exerting a biasing force against the translating element in a direction of the expandable element; wherein, upon contraction of the expandable element, the biasing force of the spring is directed to cause a linear displacement of the translating element toward the expandable element and a resulting inward rotation of the rigid support petals by the linkage members.
9. The well interval isolation device of claim 6, wherein the retraction mechanism comprises a motive device selected from the group consisting of a compression spring, a leaf spring, and a linear actuator, the motive device directly pivotably coupled between the body and second ends of the rigid support petals.
10. The well interval isolation device of claim 6, wherein the retraction mechanism encircles and is in contact with the rigid support petals and comprises a device selected from the group consisting of an elastomeric band, an elastomeric ring, and an annular spring.
11. The well interval isolation device of claim 6, wherein: the uphole rigid support element is arranged on an uphole side of the uphole expandable element to resist an uphole-directed deformation of the uphole expandable element; and the downhole rigid support element is arranged on a downhole side of the downhole expandable element to resist a downhole-directed deformation

of the downhole expandable element.

12. The well interval isolation device of claim 11, wherein: the annular array of overlapping and pivotable rigid support petals of the uphole rigid support element are adjacent to and encircle an uphole end of the uphole expandable element; and the annular array of overlapping and pivotable rigid support petals of the downhole rigid support element are adjacent to and encircle a downhole end of the downhole expandable element.

13. The well interval isolation device of claim 6, wherein: the uphole rigid support element is arranged on a downhole side of the uphole expandable element to resist a downhole-directed deformation of the uphole expandable element; and the downhole rigid support element is arranged on an uphole side of the downhole expandable element to resist an uphole-directed deformation of the downhole expandable element.

14. The well interval isolation device of claim 13, wherein: the annular array of overlapping and pivotable rigid support petals of the uphole rigid support element are adjacent to and encircle a downhole end of the uphole expandable element; and the annular array of overlapping and pivotable rigid support petals of the downhole rigid support element are adjacent to and encircle an uphole end of the downhole expandable element.

15. A method comprising: positioning a well interval isolation device at a target location within an open-hole portion of a wellbore, the well interval isolation device comprising: a downhole tool; and an uphole expandable element isolation assembly and a downhole expandable element isolation assembly coupled to the downhole tool near opposite ends thereof, each expandable element isolation assembly comprising: an expandable element; at least one rigid support element encircling at least a part of the expandable element such that a portion of the at least one rigid support element is positionable between the at least a part of the expandable element and an inner wall of a formation at an open-hole portion of the wellbore in response to expansion of the expandable element; and a retraction mechanism for retracting the at least one rigid support element upon contraction of the expandable element; and isolating an interval of the wellbore by expanding the expandable elements such that an outside surface of each expandable element is forced into contact with the inner wall of the formation and the at least one rigid support element associated with each expandable element is positioned between at least part of the expandable element and the inner wall of the formation by expansion of the expandable element.

16. The method of claim 15, wherein the retraction mechanism comprises: a plurality of linkage members pivotably connected at first ends thereof to second ends of corresponding ones of an annular array of overlapping and pivotable rigid support petals, second ends of the plurality of linkage members pivotably connected to a translating element that is coupled to a body so as to be linearly displaceable relative thereto; and a spring exerting a biasing force against the translating element in a direction of the expandable element; wherein, upon contraction of the expandable element, the biasing force of the spring causes a linear displacement of the translating element toward the expandable element, which produces an inward rotation of the rigid support petals by the linkage members.

17. The method of claim 16, wherein the retraction mechanism comprises a motive device selected from the group consisting of a compression spring, a leaf spring, and a linear actuator, the motive device directly pivotably coupled between the body and second ends of the rigid support petals.

18. The method of claim 16, wherein the retraction mechanism encircles and is in contact with the rigid support petals and comprises a device selected from the group consisting of an elastomeric band, an elastomeric ring, and an annular spring.

19. The method of claim 15, wherein the expandable elements are expanded by inflation with a pressurized fluid.
